

RECSYS CHALLENGE 2026 · MUSIC-CRS

# State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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11

retrieval branches

146

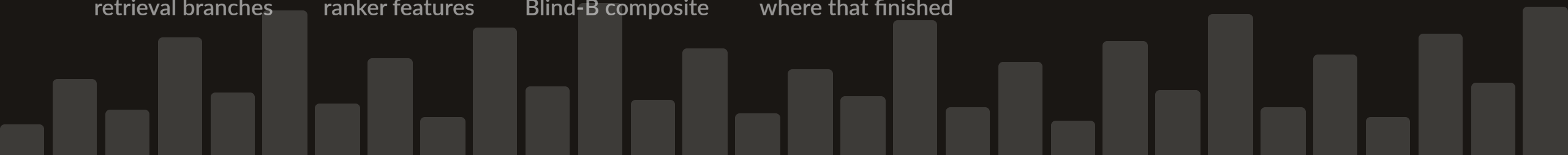
ranker features

0.3811

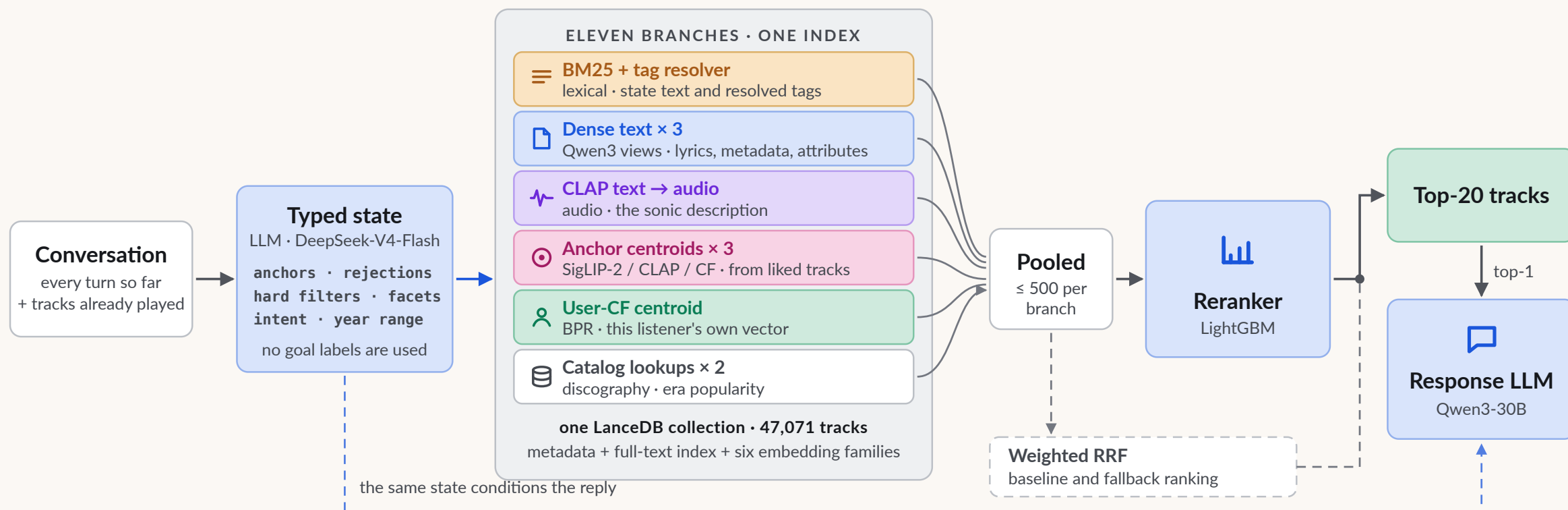
Blind-B composite

29/40

where that finished



# One state, eleven branches, one learned ranker





# 4 Re-ranking

one learned model over the pooled candidates



# What the ranker leans on

| FEATURE FAMILY              | #  | GAIN  |
|-----------------------------|----|-------|
| Bi-encoder cosine           | 1  | 59.0% |
| Per-branch rank and score   | 66 | 11.5% |
| Session and artist affinity | 18 | 10.7% |
| Popularity · state · tags   | 31 | 9.2%  |
| Other similarity cosines    | 17 | 8.9%  |
| Constraints and rejections  | 13 | 0.7%  |

## SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

## READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

## GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.



5

# Response generation

writing the reply — 30% of the composite



# Four ways the reply went wrong

1

**“Got it — here's *Alfie*...”**

for a track we do not have

CLAIMS COMPLIANCE

2

**“Sorry, this doesn't match your request.”**

about our own rank-1 pick

ARGUES AGAINST ITSELF

3

**Asked a question, given a recommendation**

“is it the rhythm or the instrumentation?”

DODGES

4

**“hauntingly beautiful... raw emotion”**

no evidence from the track or the conversation

GENERIC

Rows 2 and 3 are the prompt **working as written** — it was told to flag a conflict rather than oversell.



# 6 What it scored

the official splits, and what we steered on





# How it scored

| OFFICIAL CODABENCH SPLIT    | NDCG@20 | CAT.   | LEX.   | JUDGE | COMPOSITE |
|-----------------------------|---------|--------|--------|-------|-----------|
| Blind-A · development phase | 0.4380  | 0.0313 | 0.8028 | 4.70  | 0.5799    |
| Blind-B · final, 29th of 40 | 0.2537  | 0.0315 | 0.7862 | 3.30  | 0.3811    |

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

category and lexical barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



# 7 What went wrong

four findings, and the turns behind them



# What is not ideal

WENT WRONG

CORE

1

## We tuned on what we measured

The development set was both our training and our evaluation data.

2

## We extracted more than we used

Most fields in the state never reached retrieval or ranking at all.

3

## Nothing from outside the catalog

No tempo, key or soundtrack. Past one CF vector, no listening behaviour.

4

## The reply was never checked

It described a substitute as though it were the track asked for.



# Asked for, served

WENT WRONG

CORE

1

*“the Kamelot track from **Silverthorn**”*

**Kamelot — Fallen Star**

from *Haven*. *Silverthorn* is in the catalog.

CONSTRAINT

2

*“exactly **126.70 bpm**, in *G minor*”*

**no tempo or key field exists**

nothing in the catalog could match it

NO DATA

3

*“**Watercolors**’ by *Pat Metheny*”*

**Pat Metheny — Alfie**

described as if it were the track asked for

RESPONSE



# Extracted, then ignored

WENT WRONG

CORE



*"I gotta break away from **Deltron** and **Del** completely"*

**Dead Prez — Hip Hop**

state held both artists as avoid

IGNORED



*"branching out to **other** singer-songwriter artists"*

**Ryan Adams — When The Stars Go Blue**

state held Ryan Adams as avoid

IGNORED



*"still focusing on **Juanes**... haven't found the depth I'm after"*

**Juanes — Me Enamorado De Ti**

state held Juanes as rejected

IGNORED

Three sessions, one failure: the listener asked to move on, the state recorded it, and **the ranker served that artist anyway.**



# What the training labels look like

WENT WRONG

CORE



*"I keep asking for **Arcade Fire** — that's the only song I want"*

label: **Arctic Monkeys — This House is a Circus**

DOES NOT



*"similar chill vibe, but **from a different artist**"*

label: **Bonobo — Jets**

MOVES



*"what **other artists** fit this darkwave style?"*

label: **Tropic of Cancer — The Dull Age**

MOVES



*"what **other** orchestral tracks would you recommend?"*

label: **Nick Phoenix — Kogan**

MOVES

Three of four are annotated **MOVES TOWARD GOAL** — the dataset records them as having helped. Train on this and the model learns to anchor. **Ours does.**



# The target often ignores the request

ANALYSIS

CORE

**User:** "...something with a similar chill, electronic vibe, but **from a different artist?**" (just played: Bonobo)

**Target:** **Bonobo – "Jets"** · annotated MOVES toward goal

**Next turn:** "'Jets' is cool, but I was actually hoping for something **from a different artist this time...**"

**Target:** **Bonobo – "Pieces"** · annotated MOVES toward goal

Ground truth is simply **whatever played next**, in a session an LLM wrote. Our own model anchors on the just-played artist — we think it learns that here.



# 8 Where it leaves us

what we would change, and what is released



# What we would do differently

CONCLUSION

CORE

- 1 **Hold out a clean split before touching anything.** Free, and it would have changed every selection decision we made.
- 2 **Make the state executable.** Rejections and constraints carried 0.7% of model gain — most of what the extractor understood never became a filter.
- 3 **Import knowledge the catalog lacks** — tempo, key, soundtrack and film/TV links — so requests naming them have something to match.
- 4 **Check the response against the state** before emitting it: flag unavailable tracks and unmet constraints instead of overselling a substitute.
- 5 **Decide what to do when the benchmark and your judgment disagree.** We still do not have a principled answer.



# Everything is released

## IN THE REPO

- Pipeline code, configs, trained models
- Cached per-turn extraction states
- The 113,393 re-judged labels
- A frozen-LLM-cache replay bundle

## ZERO-CREDENTIAL REPLAY

Every LLM call is served from a curated frozen cache that **hard-fails on a miss** rather than falling through to a paid call. Both submissions reproduce without any API key.



**Code, models, data**

[github.com/npatta01/  
music-crs-2026](https://github.com/npatta01/music-crs-2026)